

Christopher Annunziato

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EDUCATION

MSPH, Biostatistics (Data Science Certificate)

Emory University, Atlanta, GA

Expected May 2026 • GPA: 3.74

BS, Statistics (Applied Computing Minor)

Arizona State University, Tempe, AZ

May 2024 • GPA: 3.61

TECHNICAL SKILLS

Statistical Tools: R, SAS, Python

Data Tools: SQL, Snowflake, Tableau, Excel

Programming: MATLAB, C++, C, Java

Methods: GLMs, mixed-effects models, survival analysis, cox proportional hazards, causal inference, machine learning, Bayesian methods

Reproducibility & Workflow: Git/GitHub, Docker, GNU Make, RMarkdown, Quarto

RELEVANT COURSEWORK (GRADUATE)

- **Data Science Toolkit (DATA 550):** Reproducible pipelines using command line tools, Git/GitHub, Docker, RMarkdown, and Makefiles.
- **Survival Analysis (BIOS 522):** Kaplan-Meier estimation, log-rank tests, Cox proportional hazards models, parametric survival modeling, censoring mechanisms.
- **Modern Regression (BIOS 526):** Mixed models, generalized estimation equations (GEEs), generalized linear models (GLMs), generalized additive models (GAMs), principal component analysis (PCA), introductory Bayesian modeling.
- **Machine Learning (BIOS 534):** Basic machine learning theory, classification methods, model generalization, clustering, dimension reduction.

- **Applied Linear Models (BIOS 509):** Statistical methodology for continuous outcome data, secondary hypothesis testing, regression diagnostics, model selection, confounding, interaction, random effect modeling.

Additional coursework: Longitudinal Data Analysis, Clinical Trials, Causal Inference, Geographic Information Systems (GIS).

INDUSTRY & RESEARCH EXPERIENCE

American Cancer Society — Research Intern

April 2025–October 2025 (20 hours/week)

PI: Parichoy Pal Choudhury, PhD

- Worked under the supervision of Dr. Parichoy Pal Choudhury on epidemiologic and biostatistical cancer research.
- Analyzed Surveillance, Epidemiology, and End Results (SEER) breast cancer data to support epidemiologic research.
- Developed a historical baseline Cox proportional hazard models to evaluate treatment outcomes.
- Applied propensity score matching/weighting to build adjusted survival predictors.
- Created linear predictors for analyzing treatment outcomes by age groups.
- Demonstrated possible alternative analysis methods to risk prediction team.
- Attended weekly risk prediction team meetings to discuss current projects and provide feedback

Drivetime — Supply Chain Analytics Intern

June 2023–August 2023 (40 hours)

- Queried large-scale datasets using Snowflake to support pricing, purchasing, and supply chain decisions.
- Designed and prototyped a key performance indicator (KPI) goals-tracking dashboard and presented the output and capabilities to department directors.
- Created a dashboard to determine the timeliness of car auctions throughout the country to determine where to target purchases.
- Analyzed effectiveness of in-house sales predicting model and produced feedback on limitations and improvements.

- Produced simulation of call center workflow to determine most effective scheduling system.
- Delivered bi-weekly analytics projects and data visualizations using Tableau.

Biostatistics Capstone — Graduate Research Project

Fall 2025–Spring 2026

PI: Robert Lyles PhD

Conducting graduate research under the supervision of Dr. Robert Lyles, Professor in the Department of Biostatistics and Bioinformatics at Emory University's Rollins School of Public Health, focused on population-average inference for clustered binary outcomes under nonlinear mixed models.

- Investigating probit random-intercept models for repeated binary outcomes, with emphasis on how the probit link induces natural variation in the population average (PA) log odds ratio as a function of the continuous exposure.
- Studying the relationship between subject-specific and population-average interpretations under the probit-normal model, contrasting implications with logistic-normal mixed models and unweighted GEE approaches.
- Implementing subject level bootstrap procedures combined with density estimation for the exposure to estimate an overall PA log odds ratio that averages over the exposure distribution.
- Contributing simulation studies evaluating bias, efficiency, and inferential properties of PA estimators under the probit-normal model, including comparisons to alternative modeling strategies.
- Supporting methodological development and manuscript preparation through reproducible simulation code and statistical analysis.

Statistics Capstone — Undergraduate Research Project

Fall 2023

- Collaborated with ASU opioid research team to assess effects of extended emergency clinic hours on crime.
- Produced literature review and determined appropriate statistical modeling approaches.
- Used a difference in differences method to determine that there was no statistically significant difference in crime between counties with extended clinic hours and counties with no changes.

ADDITIONAL EXPERIENCE

Arizona State University Sun Devil Fitness Complex

- **Facilities Manager**

January 2023–May 2024 (20-35 hours)

Managed daily operations, supervised teams of 3-5, managed monthly community events, and handled membership sales and customer inquiries.

- **Facilities Supervisor**

May 2022–January 2023 (20-35 hours)

Supported facility oversight and safety compliance; assisted with staff coordination and customer service.

ADDITIONAL INFORMATION

References, full course list, and code samples available upon request.